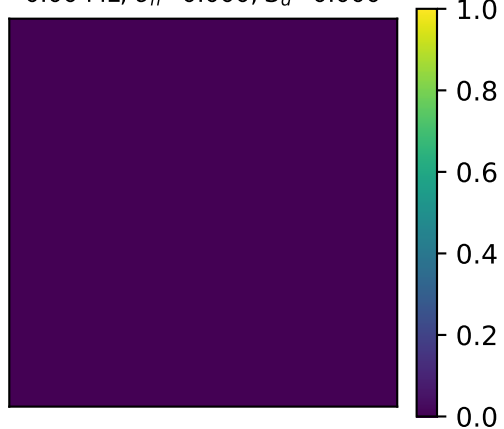


Synthetic surfaces and their kinematic screens — 1.648° grazing, 15 keV, 4.0 nm coherence.

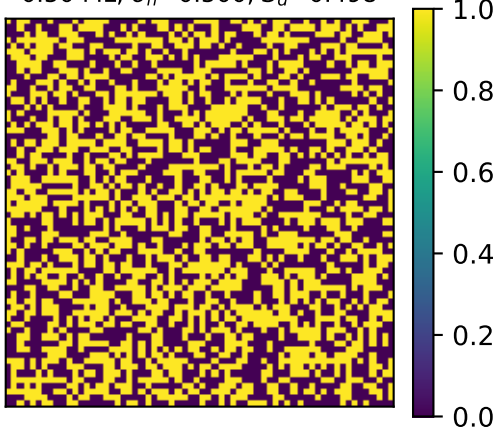
A is the reference: any departure from it is disorder.

B and C are both exactly 0.50 ML on two levels, yet only B cancels: cancellation needs both terraces inside one coherence area, and C's terraces are 10.2 nm wide against a 4 nm beam. See the coherence sweep on the last page.

A perfectly flat
0.00 ML, $\sigma_h=0.000$, $S_d=0.000$



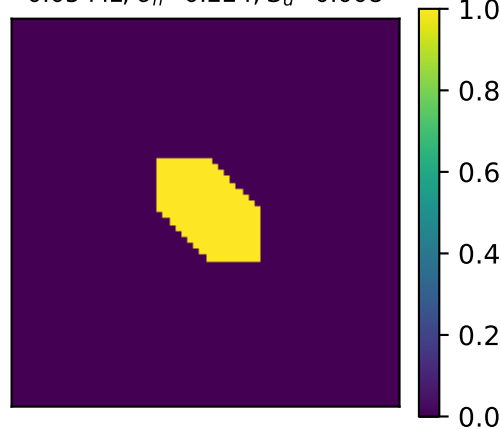
B half layer, two levels
0.50 ML, $\sigma_h=0.500$, $S_d=0.498$



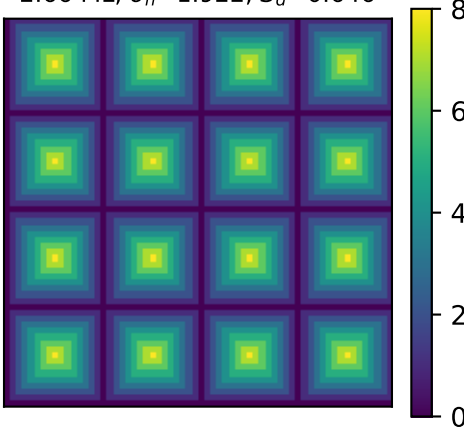
C straight step (x2, periodic)
0.50 ML, $\sigma_h=0.500$, $S_d=0.021$



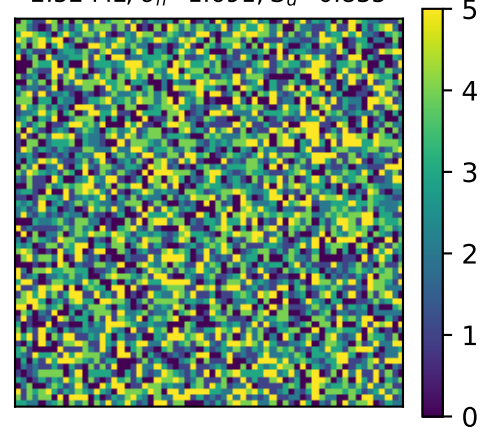
D isolated island
0.05 ML, $\sigma_h=0.224$, $S_d=0.008$



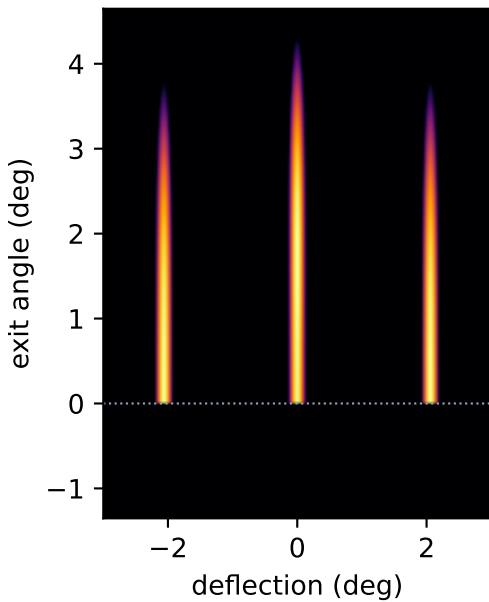
E periodic pyramid array
2.66 ML, $\sigma_h=1.922$, $S_d=0.646$



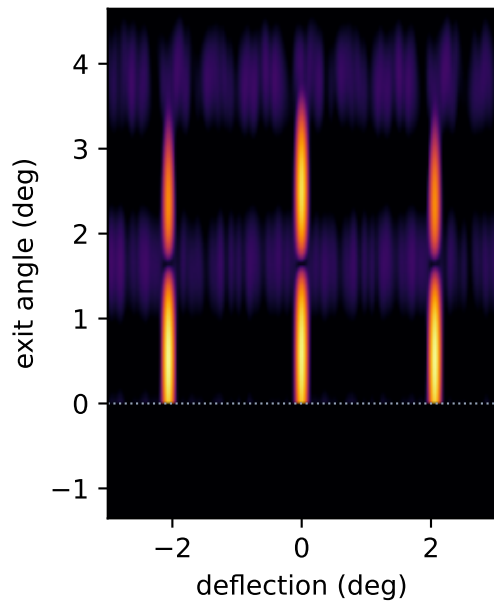
F random rough
2.52 ML, $\sigma_h=1.691$, $S_d=0.835$



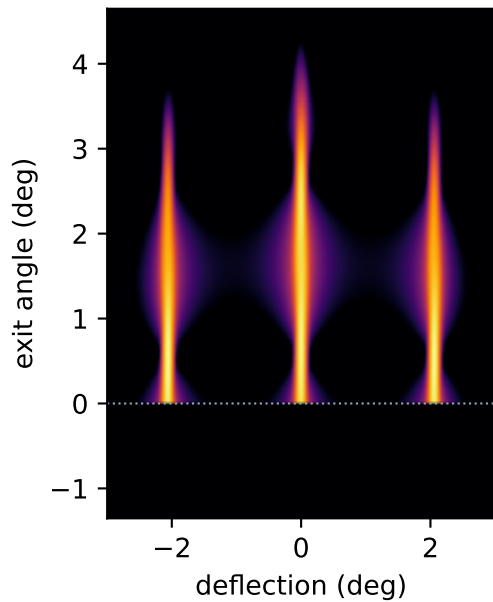
$I_{00} = 1.0000$



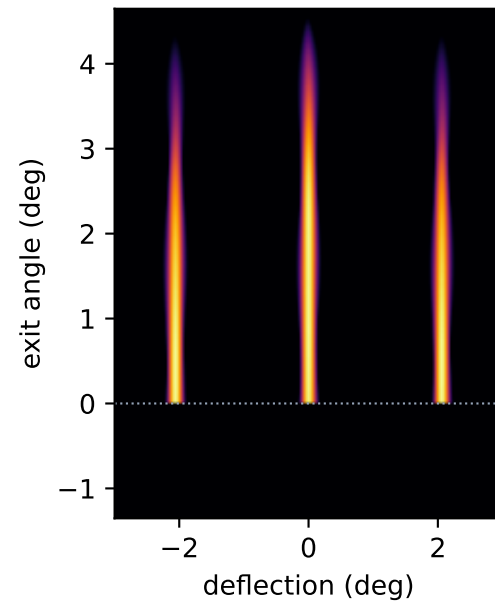
$I_{00} = 0.0012$



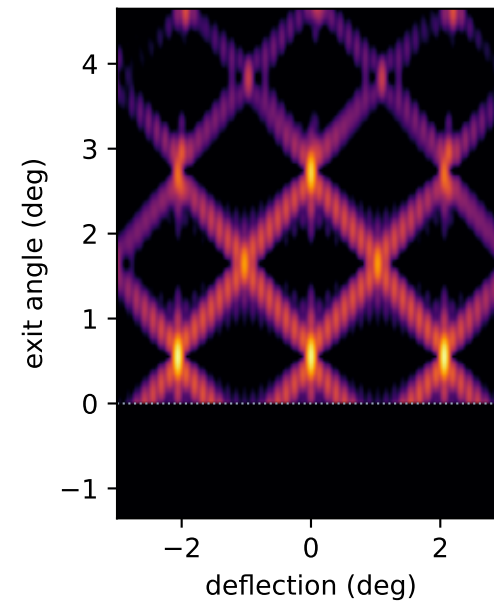
$I_{00} = 0.5487$



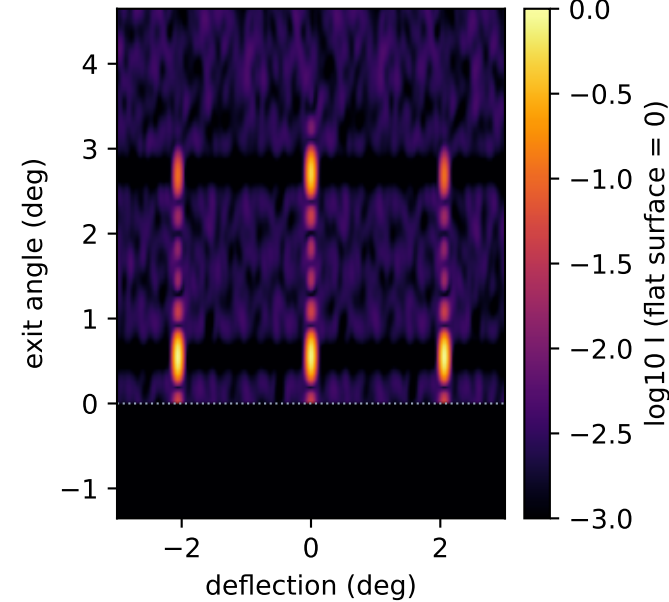
$I_{00} = 0.8464$



$I_{00} = 0.0000$

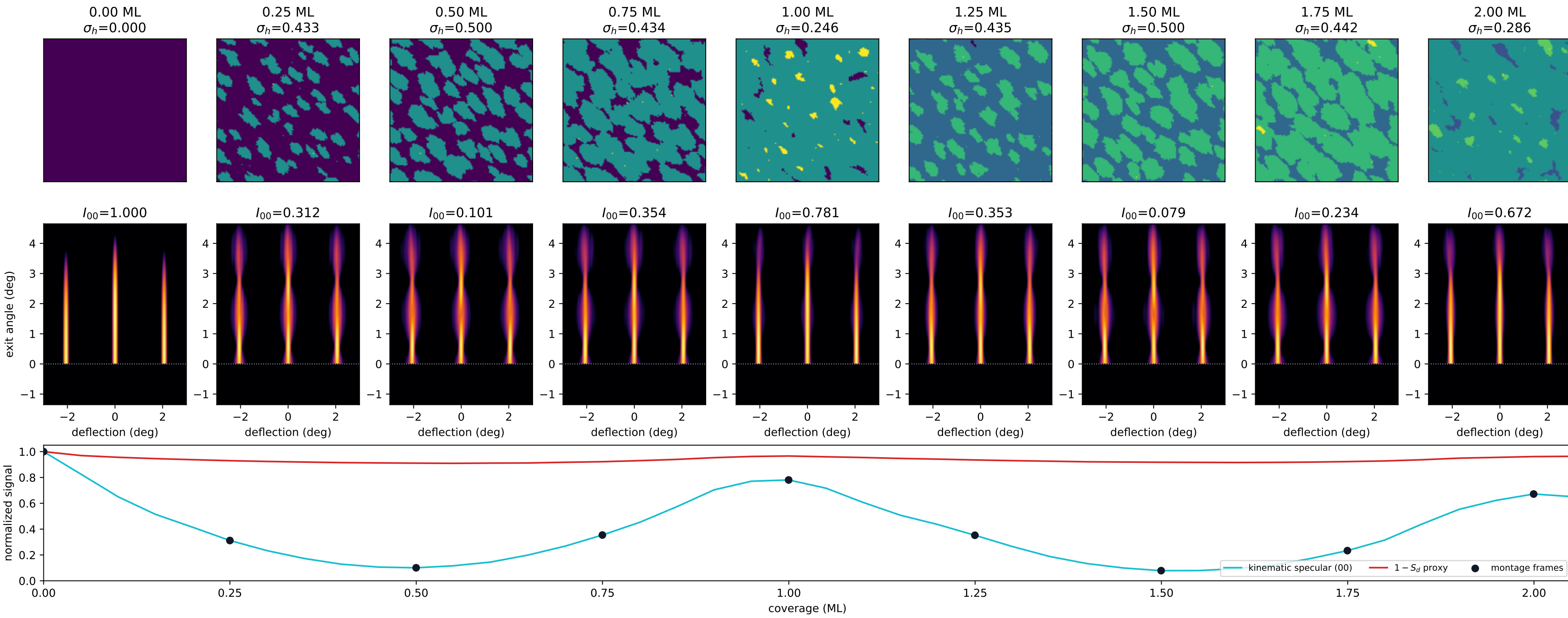


$I_{00} = 0.0037$



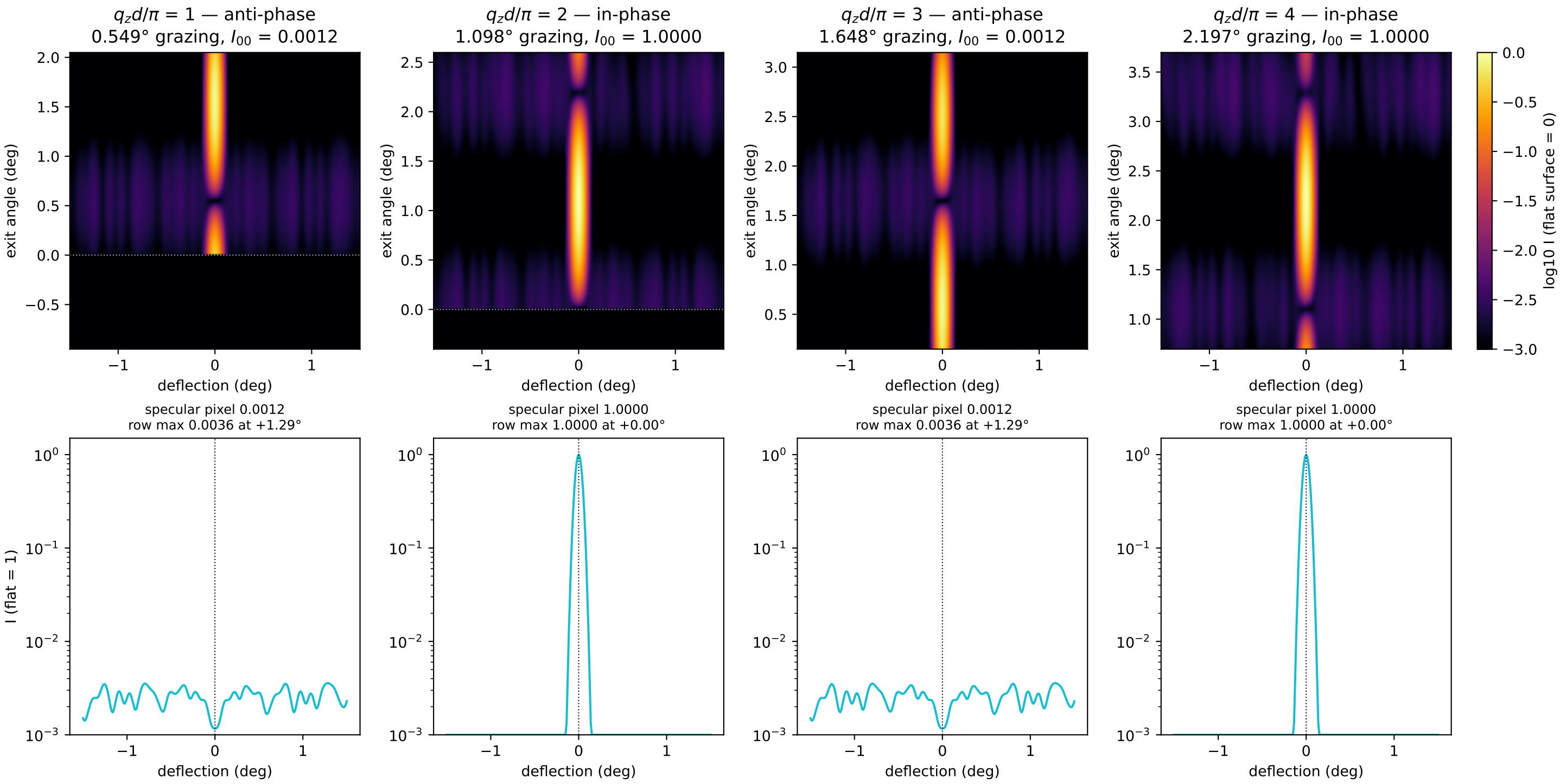
Coverage montage, 0 to 2 ML — Ga/N = 0.82 paper preset, 128x128, 1.648° grazing.

Roughens near half coverage, partially smooths near integer ML: one cycle per monolayer, but not a return to flat — at 1 ML σ_h is still 0.246 and the specular recovers only to 0.781. The incomplete recovery is the damping.



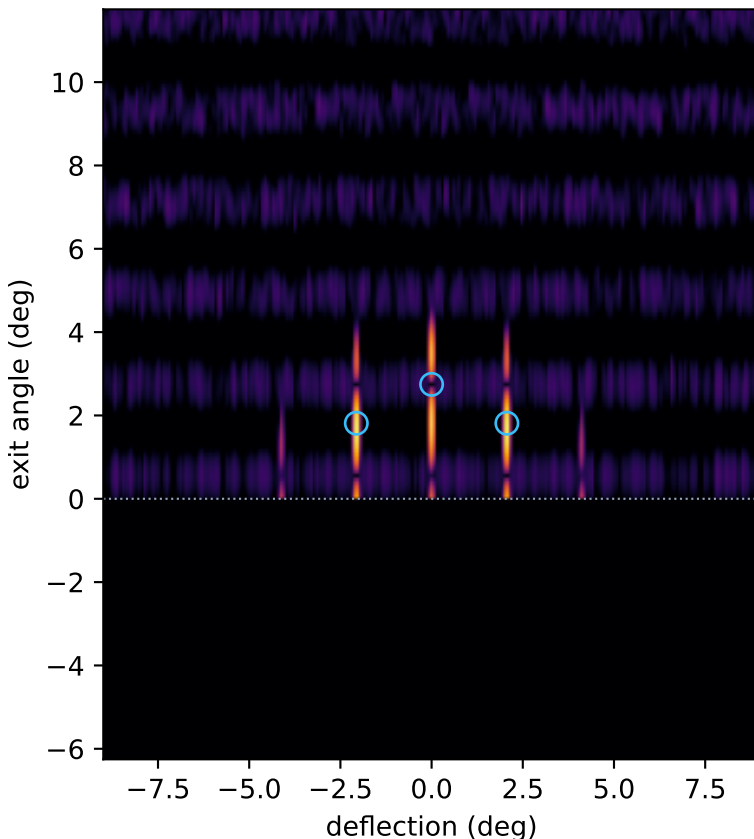
Phase-order sweep on one frozen half-covered two-level surface (0.50 ML).

Odd orders put adjacent terraces pi out of phase and must cancel; even orders must recover. The surface never changes — only the beam does.

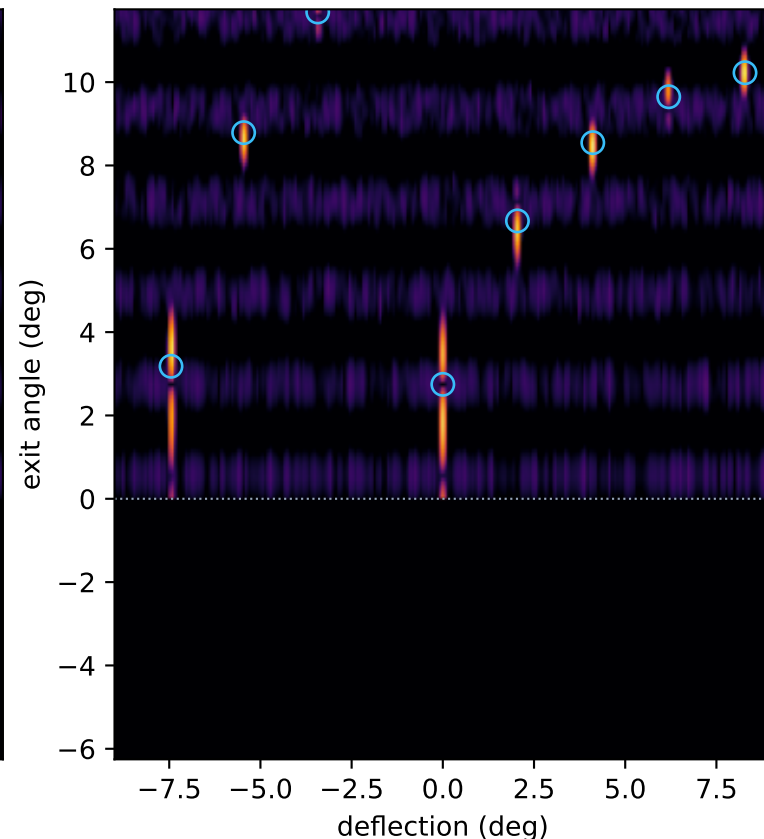


Azimuth sweep on one frozen surface — 2.747° grazing (phase order 5), where first orders exist.
Rotating the sample sweeps the Ewald sphere across a hexagonal reciprocal lattice, so which rods are reachable changes. Circles are `rod\_orders()`, not hand placement.
0° and 30° are the two inequivalent high-symmetry directions; the pattern repeats every 60° and mirrors at 30°.

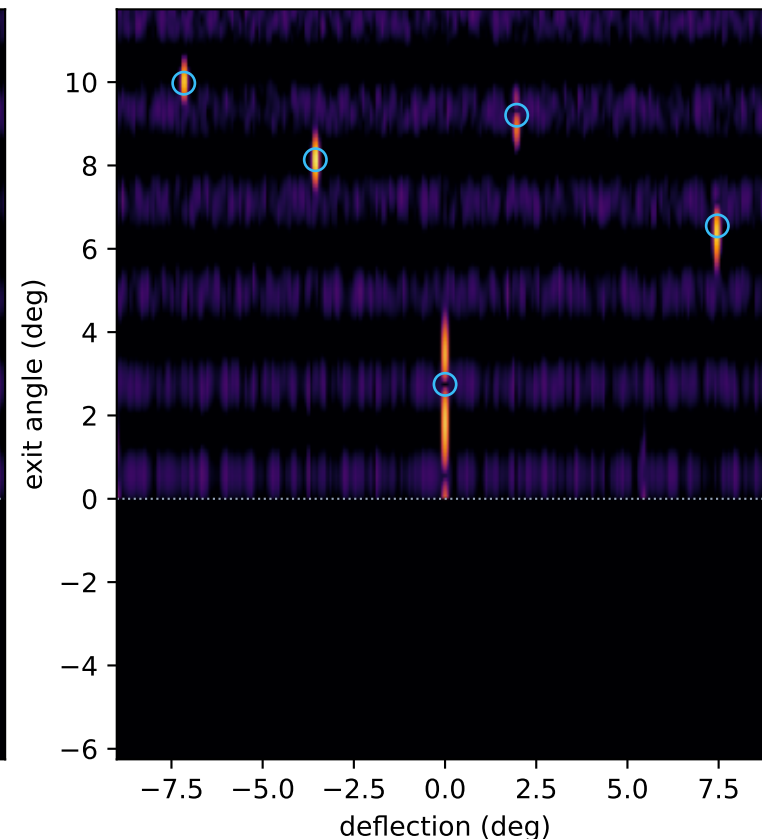
azimuth 0° — 3 reachable rods
(0,-1), (00), (01)



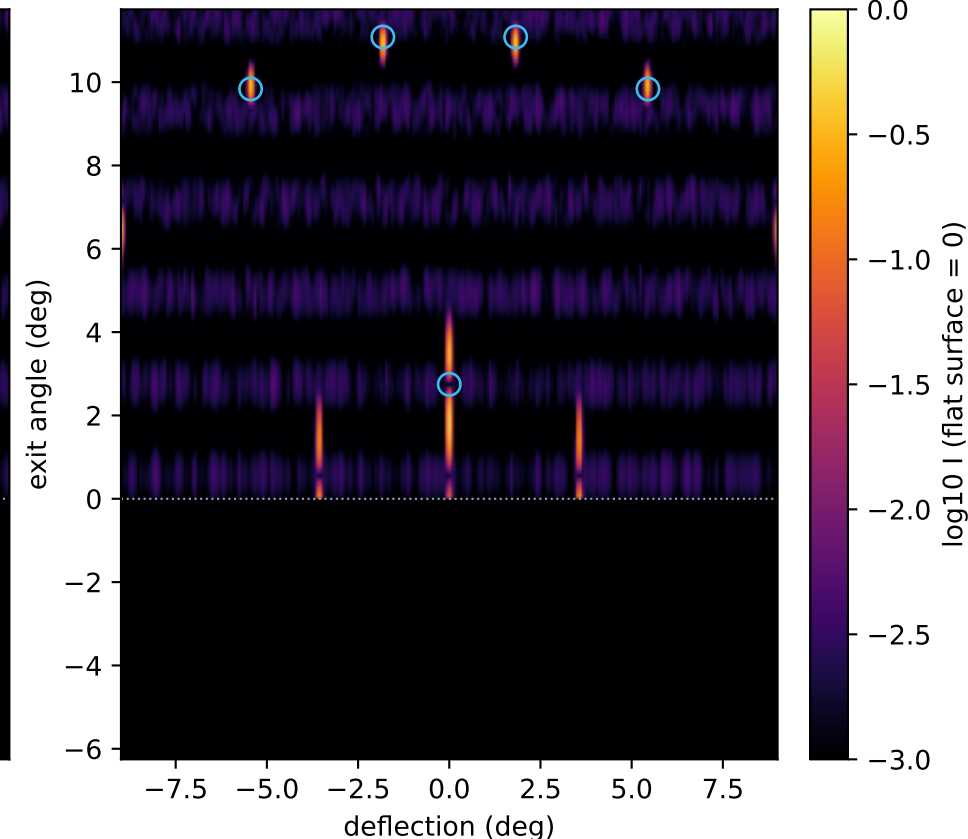
azimuth 10° — 8 reachable rods
(-1,-4), (-1,-3), (-1,-2), (00)
(01), (02), (03), (04)



azimuth 20° — 5 reachable rods
(-2,-4), (-1,-2), (00), (01)
(14)



azimuth 30° — 5 reachable rods
(-2,-3), (-1,-1), (00), (01)
(13)



Coherence sweep on one frozen stepped surface (0.50 ML, two levels) — 1.648° grazing.

Longer coherence narrows the rods and, on this surface, destroys the specular beam:

at 1 nm the beam sees one flat terrace ($I_{00} = 0.99$), at 16 nm it spans both and they cancel ($I_{00} = 0.002$). Same surface throughout.

